References

[1] M. Sharif, G. Han, W. Liu, X. Huang, Cultivating multidisciplinary research and education on gpu infrastructure for mid-south institutions at the university of memphis: Practice and challenge (2025). arXiv:2504.14786.
URL https://arxiv.org/abs/2504.14786

[2] J. A. Ratches, Review of current aided/automatic target acquisition technology for military target acquisition tasks, Optical Engineering 50 (7) (2011) 072001–072001.

[3] M. Harbaugh, Unmanned aerial systems (uas) for intelligence, surveillance, and reconnaissance (isr), State-of-the-Art-Report (SOAR), Defense Systems Information Analysis Center (DSIAC).
URL https://dsiac.org/wp-content/uploads/2018/05/UNMANNED-AERIAL-SYSTEMSUAS-FOR-INTELLIGENCE-SURVEILLANCE-AND-RECONNAISSANCE-ISR.pdf

[4] M. Patterson, A. Brescia, Operation of small sensor payloads on tactical sized unmanned air vehicles, The Aeronautical Journal 114 (1157) (2010) 427–436.

[5] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, A. Abbas, Modern trends in hyperspectral image analysis: A review, IEEE Access 6 (2018) 14118–14129.

[6] M. Paoletti, J. Haut, J. Plaza, A. Plaza, Deep learning classifiers for hyperspectral imaging: A review, ISPRS Journal of Photogrammetry and Remote Sensing 158 (2019) 279–317.

[7] J. Liu, X. Wang, Plant diseases and pests detection based on deep learning: a review, Plant Methods 17 (2021) 1–18.

[8] P. S. Teh, N. Zhang, A. B. J. Teoh, K. Chen, A survey on touch dynamics authentication in mobile devices, Computers & Security 59 (2016) 210–235.

[9] A. Lasisi, R. Ghazali, T. Herawan, Application of real-valued negative selection algorithm to improve medical diagnosis, in: Applied Computing in Medicine and Health, Elsevier, 2016, pp. 231–243.

[10] R. Nayak, U. C. Pati, S. K. Das, A comprehensive review on deep learning-based methods for video anomaly detection, Image and Vision Computing 106 (2021) 104078.

[11] D. G. Manolakis, R. B. Lockwood, T. W. Cooley, Hyperspectral imaging remote sensing: physics, sensors, and algorithms, Cambridge University Press, 2016.

[12] G. Geography, Multispectral vs hyperspectral imagery explained, https://gisgeography.com/multispectral-vs-hyperspectral-imagery-explained/,, accessed: 2024-11-23 (2014).
URL https://gisgeography.com/multispectral-vs-hyperspectral-imagery-explained/

[13] H. Su, Z. Wu, H. Zhang, Q. Du, Hyperspectral anomaly detection: A survey, IEEE Geoscience and Remote Sensing Magazine 10 (1) (2021) 64–90.

[14] I. Racetin, A. Krtalić, Systematic review of anomaly detection in hyperspectral remote sensing applications, Applied Sciences 11 (11) (2021) 4878.

[15] N. Keshava, J. F. Mustard, Spectral unmixing, IEEE signal processing magazine 19 (1) (2002) 44–57. doi:10.1109/79.974727.

[16] C.-I. Chang, Hyperspectral data processing: algorithm design and analysis, John Wiley & Sons, 2013.

[17] J. A. Richards, X. Jia, Remote sensing digital image analysis: an introduction, Springer, 2006.

[18] X. Hu, C. Xie, Z. Fan, Q. Duan, D. Zhang, L. Jiang, X. Wei, D. Hong, G. Li, X. Zeng, et al., Hyperspectral anomaly detection using deep learning: A review, Remote Sensing 14 (9) (2022) 1973. doi:10.3390/rs14091973.

[19] C.-I. Chang, S.-S. Chiang, Anomaly detection and classification for hyperspectral imagery, IEEE Transactions on geoscience and remote sensing 40 (6) (2002) 1314–1325.

[20] R. Zhao, B. Du, L. Zhang, A robust nonlinear hyperspectral anomaly detection approach, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 7 (4) (2014) 1227–1234.

[21] R. Zhao, Z. Shi, Z. Zou, Z. Zhang, Ensemble-based cascaded constrained energy minimization for hyperspectral target detection, Remote Sensing 11 (11) (2019) 1310.

[22] J. Opitz, S. Burst, Macro F1 and Macro F1, arXiv preprint arXiv:1911.03347.

[23] X. Kang, X. Zhang, S. Li, K. Li, J. Li, J. A. Benediktsson, Hyperspectral anomaly detection with attribute and edge-preserving filters, IEEE Transactions on Geoscience and Remote Sensing 55 (10) (2017) 5600–5611.

[24] B. Hosseiny, M. Mahdianpari, M. Hemati, A. Radman, F. Mohammadimanesh, J. Chanussot, Beyond supervised learning in remote sensing: A systematic review of deep learning approaches, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 17 (2023) 1035–1052.

[25] S. Liu, Z. Li, G. Wang, X. Qiu, T. Liu, J. Cao, D. Zhang, Spectral–spatial feature fusion for hyperspectral anomaly detection, Sensors 24 (5) (2024) 1652.

[26] X. Zhang, K. Gao, J. Wang, P. Wang, Z. Hu, Z. Yang, X. Zhao, W. Li, Target detection adapting to spectral variability in multi-temporal hyperspectral images using implicit contrastive learning, Remote Sensing 16 (4) (2024) 718.

[27] Y. Zhu, F. Zhuang, J. Wang, G. Ke, J. Chen, J. Bian, H. Xiong, Q. He, Deep subdomain adaptation network for image classification, ieee transactions on neural networks and learning systems 32 (2021) 1713–1722. doi:10.1109/tnnls.2020.2988928.

[28] H. Liu, R. Jiang, B. Zhou, An ensemble approach for unsupervised anomaly and cyber attack detection, journal of computer security 33 (2025) 215–238. doi:10.1177/0926227x251330216.

[29] S. Nazat, W. Alayed, L. Li, M. Abdallah, Ensemble learning framework for anomaly detection in autonomous driving systems, Sensors 25 (16) (2025) 5105.

[30] M. S. Younis, M. Hossain, A. L. Robinson, L. Wang, C. Preza, Hyperspectral unmixing-based anomaly detection, in: Computational Imaging VII, Vol. 12523, SPIE, 2023, p. 1252302. doi:10.1117/12.2664706.

[31] M. Hossain, M. Younis, A. Robinson, L. Wang, C. Preza, Greedy ensemble hyperspectral anomaly detection, Journal of Imaging 10 (6) (2024) 131. doi:10.3390/jimaging10060131.

[32] M. Hossain, A. Robinson, L. Wang, C. Preza, Investigation of unsupervised and supervised hyperspectral anomaly detection, in: Applications of Machine Learning 2024, Vol. 13138, SPIE, 2024, p. 1313817. doi:10.1117/12.3029916.

[33] M. Hossain, A. Robinson, L. Wang, C. Preza, Improving semantic segmentation through task adaptation for uav hyperspectral agricultural imagery, in: Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping X, Vol. 13475, SPIE, 2025, p. 1347507. doi:10.1117/12.3053426.

[34] M. Hossain, A. Robinson, C. Preza, L. Wang, Gmoe-ad: Generalized hyperspectral anomaly detection via mixture-of-experts and domain-invariant learning, Sensors 26 (2026) 5661. doi:10.3390/s26175661.

[35] H. Kwon, N. M. Nasrabadi, Kernel rx-algorithm: A nonlinear anomaly detector for hyperspectral imagery, IEEE Transactions on Geoscience and Remote Sensing 43 (2) (2005) 388–397.

[36] N. Acito, M. Diani, G. Corsini, Gaussian mixture model based approach to anomaly detection in multi/hyperspectral images, in: Image and Signal Processing for Remote Sensing XI, Vol. 5982, SPIE, 2005, pp. 209–217.

[37] A. Schaum, Joint subspace detection of hyperspectral targets, in: 2004 IEEE Aerospace Conference Proceedings (IEEE Cat. No. 04TH8720), Vol. 3, IEEE, 2004, p. 1824.

[38] S. Li, K. Zhang, P. Duan, X. Kang, Hyperspectral anomaly detection with kernel isolation forest, IEEE Transactions on Geoscience and Remote Sensing 58 (1) (2019) 319–329.

[39] K. Tan, Z. Hou, F. Wu, Q. Du, Y. Chen, Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation, Remote sensing 11 (11) (2019) 1318.

[40] J. Sill, G. Takács, L. Mackey, D. Lin, Feature-weighted linear stacking, arXiv preprint arXiv:0911.0460.

[41] C.-I. Chang, An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis, IEEE Transactions on Information Theory 46 (5) (2000) 1927–1932. doi:10.1109/18.857802.

[42] B. Niu, Q. Feng, B. Chen, C. Ou, Y. Liu, J. Yang, Hsi-transunet: A transformer based semantic segmentation model for crop mapping from uav hyperspectral imagery, Computers and Electronics in Agriculture 201 (2022) 107297.

[43] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, Y. Zhou, TransUNet: Transformers make strong encoders for medical image segmentation, arXiv preprint arXiv:2102.04306 1.

[44] T. P. Watson, K. McKenzie, A. Robinson, K. Renshaw, R. Driggers, E. L. Jacobs, J. Conroy, Evaluation of aerial real-time rx anomaly detection, in: Algorithms, Technologies, and Applications for Multispectral and Hyperspectral Imaging XXIX, Vol. 12519, SPIE, 2023, pp. 254–260.

[45] M. K. Griffin, H.-h. K. Burke, Compensation of hyperspectral data for atmospheric effects, Lincoln laboratory journal 14 (1) (2003) 29–54.

[46] G. P. Anderson, G. W. Felde, M. L. Hoke, A. J. Ratkowski, T. W. Cooley, J. H. Chetwynd Jr, J. Gardner, S. M. Adler-Golden, M. W. Matthew, A. Berk, et al., Modtran4-based atmospheric correction algorithm: Flaash (fast line-of-sight atmospheric analysis of spectral hypercubes), in: Algorithms and technologies for multispectral, hyperspectral, and ultraspectral imagery VIII, Vol. 4725, SPIE, 2002, pp. 65–71.

[47] T. Wang, T. Guan, F. Qiu, L. Liu, X. Zhang, H. Zeng, Q. Zhang, Evaluation of scale effects on uav-based hyperspectral imaging for remote sensing of vegetation, Remote Sensing 17 (6) (2025) 1080.

[48] A. Maity, A. Pattanaik, S. Sagnika, S. Pani, A comparative study on approaches to speckle noise reduction in images, in: 2015 International Conference on Computational Intelligence and Networks, IEEE, 2015, pp. 148–155.

[49] R. C. Gonzalez, R. E. Woods, Digital image processing, Pearson education india, 2009.

[50] K. Dabov, A. Foi, V. Katkovnik, K. Egiazarian, Image denoising by sparse 3-d transform-domain collaborative filtering, IEEE Transactions on image processing 16 (8) (2007) 2080–2095.

[51] Y. Mäkinen, L. Azzari, A. Foi, Collaborative filtering of correlated noise: Exact transform-domain variance for improved shrinkage and patch matching, IEEE Transactions on Image Processing 29 (2020) 8339–8354.

[52] A. A. Green, M. Berman, P. Switzer, M. D. Craig, A transformation for ordering multispectral data in terms of image quality with implications for noise removal, IEEE Transactions on geoscience and remote sensing 26 (1) (1988) 65–74.

[53] C. C. Aggarwal, Outlier ensembles: position paper, ACM SIGKDD Explorations Newsletter 14 (2) (2013) 49–58.

[54] A. Zimek, R. J. Campello, J. Sander, Ensembles for unsupervised outlier detection: challenges and research questions a position paper, Acm Sigkdd Explorations Newsletter 15 (1) (2014) 11–22.

[55] T. Välikangas, T. Suomi, L. L. Elo, A systematic evaluation of normalization methods in quantitative label-free proteomics, Briefings in bioinformatics 19 (1) (2018) 1–11.

[56] Y. Zhao, L. Wong, W. W. B. Goh, How to do quantile normalization correctly for gene expression data analyses, Scientific reports 10 (1) (2020) 15534.

[57] R. Farmer, Data Transformation.
URL https://dataalltheway.com/posts/001-data-transformation

[58] S. C. Hicks, R. A. Irizarry, When to use quantile normalization?, BioRxiv (2014) 012203.

[59] A. Ghadiri, Implementation of an automated image processing system for observing the activities of honey bees, Ph.D. thesis, Appalachian State University Boone, NC, USA (2013).

[60] M. Abdullah-Al-Wadud, M. H. Kabir, M. A. A. Dewan, O. Chae, A dynamic histogram equalization for image contrast enhancement, IEEE transactions on consumer electronics 53 (2) (2007) 593–600.

[61] D. T. Susetianingtias, S. Madenda, D. A. Rahayu, et al., Retinal microaneursym detection using maximally stable external region algorithm, International Journal on Advanced Science, Engineering and Information Technology 6 (5) (2016) 644–648.

[62] I. The MathWorks, Contrast limited adaptive histogram equalization, https://www.mathworks.com/help/visionhdl/ug/contrast-adaptive-histogram-equalization.html, , accessed: 2024-12-23 (2024).
URL https://www.mathworks.com/help/visionhdl/ug/contrast-adaptive-histogram-equalization.html

[63] K. Zuiderveld, Contrast limited adaptive histogram equalization, Academic Press Professional, Inc., USA, 1994, Ch. 45, p. 474–485.

[64] H. R. Sheikh, M. F. Sabir, A. C. Bovik, A statistical evaluation of recent full reference image quality assessment algorithms, IEEE Transactions on image processing 15 (11) (2006) 3440–3451.

[65] S. Athar, Z. Wang, A comprehensive performance evaluation of image quality assessment algorithms, Ieee Access 7 (2019) 140030–140070.

[66] I. S. Reed, X. Yu, Adaptive multiple-band cfar detection of an optical pattern with unknown spectral distribution, IEEE Transactions on acoustics, speech, and signal processing 38 (10) (1990) 1760–1770.

[67] P. C. Mahalanobis, On the generalized distance in statistics, Sankhyā: The Indian Journal of Statistics, Series A (2008-) 80 (1) (2018) S1–S7.

[68] A. P. Schaum, A. D. Stocker, Joint hyperspectral subspace detection derived from a bayesian likelihood ratio test, in: Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery VIII, Vol. 4725, SPIE, 2002, pp. 225–233.

[69] A. Schaum, Hyperspectral anomaly detection beyond rx, in: Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XIII, Vol. 6565, SPIE, 2007, pp. 13–25.

[70] M. J. Carlotto, A cluster-based approach for detecting man-made objects and changes in imagery, IEEE Transactions on Geoscience and Remote Sensing 43 (2) (2005) 374–387.

[71] P. C. Hytla, R. C. Hardie, M. T. Eismann, J. Meola, Anomaly detection in hyperspectral imagery: comparison of methods using diurnal and seasonal data, Journal of Applied Remote Sensing 3 (1) (2009) 033546.

[72] B. Du, R. Zhao, L. Zhang, L. Zhang, A spectral-spatial based local summation anomaly detection method for hyperspectral images, Signal Processing 124 (2016) 115–131.

[73] F. T. Liu, K. M. Ting, Z.-H. Zhou, Isolation forest, in: 2008 eighth ieee international conference on data mining, IEEE, 2008, pp. 413–422.

[74] D. Hong, N. Yokoya, J. Chanussot, X. X. Zhu, An augmented linear mixing model to address spectral variability for hyperspectral unmixing, IEEE Transactions on Image Processing 28 (4) (2018) 1923–1938.

[75] C.-I. Chang, Q. Du, Estimation of number of spectrally distinct signal sources in hyperspectral imagery, IEEE Transactions on Geoscience and Remote Sensing 42 (3) (2004) 608–619. doi:10.1109/TGRS.2003.819189.

[76] M. E. Winter, N-findr: an algorithm for fast autonomous spectral end-member determination in hyperspectral data, in: M. R. Descour, S. S. Shen (Eds.), Imaging Spectrometry V, Vol. 3753, International Society for Optics and Photonics, SPIE, 1999, pp. 266 – 275. doi:10.1117/12.366289.
URL https://doi.org/10.1117/12.366289

[77] Y. Qu, R. Guo, W. Wang, H. Qi, B. Ayhan, C. Kwan, S. Vance, Anomaly detection in hyperspectral images through spectral unmixing and low rank decomposition, in: 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), IEEE, 2016, pp. 1855–1858.

[78] S. Ravel, C. Fossati, S. Bourennane, Spectral unmixing of hyperspectral images in the presence of small targets, Remote Sensing 7 (1).

[79] W. Shang, M. Jouni, Z. Wu, Y. Xu, M. Dalla Mura, Z. Wei, Hyperspectral anomaly detection based on regularized background abundance tensor decomposition, Remote Sensing 15 (6) (2023) 1679. doi:10.3390/rs15061679.

[80] J. Zhang, X. Zhang, L. Jiao, Dual-view hyperspectral anomaly detection via spatial consistency and spectral unmixing, Remote Sensing 15 (13) (2023) 3330.

[81] C. Chenot, M. Yaghoobi, M. E. Davies, Y. Altmann, Anomaly detection with high resolution hyperspectral observations, in: 2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP), IEEE, 2018, pp. 963–967. doi:10.1109/GlobalSIP.2018.8646705.

[82] K. Jiang, W. Xie, J. Lei, T. Jiang, Y. Li, Lren: Low-rank embedded network for sample-free hyperspectral anomaly detection, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 35, 2021, pp. 4139–4146. doi:10.1609/aaai.v35i5.16536.

[83] X. Shang, Y. Duan, X. Wang, B. Fu, X. Sun, Anomaly-background separation and particle swarm optimization based band selection for hyperspectral anomaly detection, IET Image Processing 18 (8) (2024) 2053–2063.

[84] Y. Lu, X. Zheng, H. Xin, H. Tang, R. Wang, F. Nie, Ensemble and random collaborative representation-based anomaly detector for hyperspectral imagery, Signal Processing 204 (2023) 108835.

[85] N. Merrill, C. C. Olson, Unsupervised ensemble-kernel principal component analysis for hyperspectral anomaly detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops, 2020, pp. 112–113.

[86] X. Yang, X. Huang, M. Zhu, S. Xu, Y. Liu, Ensemble and random rx with multiple features anomaly detector for hyperspectral image, IEEE Geoscience and Remote Sensing Letters 19 (2022) 1–5.

[87] S. Wang, W. Feng, Y. Quan, W. Bao, G. Dauphin, L. Gao, X. Zhong, M. Xing, Subfeature ensemble-based hyperspectral anomaly detection algorithm, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 15 (2022) 5943–5952.

[88] S. Fatemifar, M. Awais, A. Akbari, J. Kittler, A stacking ensemble for anomaly based client-specific face spoofing detection, in: 2020 IEEE International Conference on Image Processing (ICIP), IEEE, 2020, pp. 1371–1375.

[89] J. Nalepa, M. Myller, L. Tulczyjew, M. Kawulok, Deep ensembles for hyperspectral image data classification and unmixing, Remote Sensing 13 (20) (2021) 4133.

[90] X. Zhang, H. Zhao, Hyperspectral-cube-based mobile face recognition: A comprehensive review, Information Fusion 74 (2021) 132–150.

[91] G. Seni, J. Elder, Ensemble methods in data mining: improving accuracy through combining predictions, Morgan & Claypool Publishers, 2010.

[92] C. C. Aggarwal, S. Sathe, Theoretical foundations and algorithms for outlier ensembles, Acm sigkdd explorations newsletter 17 (1) (2015) 24–47.

[93] C. Zhang, Y. Ma, Ensemble machine learning: methods and applications, Springer, 2012.

[94] L. Breiman, Stacked regressions, Machine learning 24 (1996) 49–64.

[95] B. Pavlyshenko, Using stacking approaches for machine learning models, in: 2018 IEEE Second International Conference on Data Stream Mining & Processing (DSMP), IEEE, 2018, pp. 255–258.

[96] P. de Tchébychef, Average values, Journal of Pure and Applied Mathematics 12 (1867) 177–184.

[97] X. Kang, Airport-Beach-Urban (ABU) Datasets, http://xudongkang.weebly.com/data-sets.html, , accessed: 2023-04-23 (2017).
URL http://xudongkang.weebly.com/data-sets.html

[98] Y.-Q. Zhao, J. Yang, Hyperspectral image denoising via sparse representation and low-rank constraint, IEEE Transactions on Geoscience and Remote Sensing 53 (1) (2014) 296–308.

[99] L. Zhu, G. Wen, Hyperspectral anomaly detection via background estimation and adaptive weighted sparse representation, Remote Sensing 10 (2) (2018) 272.

[100] M. Grana, M. A. Veganzons, B. Ayerdi, Hyperspectral remote sensing scenes, Grupo de Inteligencia Computacional (GIC), Accessed: 2024-01-13.
URL https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[101] W. Li, G. Wu, Q. Du, Transferred deep learning for anomaly detection in hyperspectral imagery, Git Hub, , accessed: 2024-01-13 (2019).
URL https://github.com/yousefan/CNND/blob/master/HYDICE%5Furban.mat

[102] X. Liu, M. Tanaka, M. Okutomi, Single-image noise level estimation for blind denoising, IEEE transactions on image processing 22 (12) (2013) 5226–5237.

[103] N. J. P. Laboratory, Aviris - airborne visible / infrared imaging spectrometer, https://aviris.jpl.nasa.gov/, , accessed: 2024-01-23 (2006).
URL https://aviris.jpl.nasa.gov/

[104] R. Basedow, E. Zalewski, Characteristics of the hydice sensor, in: Summaries of the Fifth Annual JPL Airborne Earth Science Workshop. Volume 1: AVIRIS Workshop, Vol. 1, 1995, p. 9.

[105] D. B. Delbosque, Hypercube, spectral imagery analysis software, https://www.erdc.usace.army.mil/media/fact-sheets/fact-sheet-article-view/article/610433/hypercube/, , accessed: 2024-04-22 (July 2015).
URL https://www.erdc.usace.army.mil/media/fact-sheets/fact-sheet-article-view/article/610433/hypercube/

[106] Resonon, Pika l - hyperspectral sensors - lightweight, compact vnir - benchtop systems, https://resonon.com/Pika-L, , accessed: 2023-09-13 (2022).
URL https://resonon.com/Pika-L

[107] C. C. Aggarwal, Outlier ensembles, in: Outlier Analysis, Springer, 2016, pp. 185–218.

[108] J. A. Hanley, B. J. McNeil, The meaning and use of the area under a receiver operating characteristic (roc) curve., Radiology 143 (1) (1982) 29–36.

[109] C.-I. Chang, Comprehensive analysis of receiver operating characteristic (roc) curves for hyperspectral anomaly detection, IEEE Transactions on Geoscience and Remote Sensing 60 (2022) 1–24. doi:10.1109/TGRS.2022.3211786.

[110] r. rasharmaravindrasharma, Auc-roc curve, https://www.geeksforgeeks.org/auc-roc-curve/, , accessed: 2024-01-13 (January 2022).
URL https://www.geeksforgeeks.org/auc-roc-curve/

[111] N. Chinchor, Muc-4 evaluation metrics, in: Proceedings of the 4th Conference on Message Understanding, MUC4 '92, Association for Computational Linguistics, USA, 1992, p. 22–29. doi:10.3115/1072064.1072067.
URL https://doi.org/10.3115/1072064.1072067

[112] T. Saito, M. Rehmsmeier, The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets, PloS one 10 (3) (2015) e0118432.

[113] L. Meegahapola, H. Hassoune, D. Gatica-Perez, M3bat: Unsupervised domain adaptation for multimodal mobile sensing with multi-branch adversarial training, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 8 (2) (2024) 46:7. doi:10.1145/3659591.

[114] A. Pillai, S. K. Nepal, W. Wang, M. Nemesure, M. Heinz, G. Price, D. Lekkas, A. C. Collins, T. Griffin, B. Buck, S. M. Preum, T. Cohen, N. C. Jacobson, D. Ben-Zeev, A. Campbell, Investigating generalizability of speech-based suicidal ideation detection using mobile phones, Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 7 (4). doi:10.1145/3631452.
URL https://doi.org/10.1145/3631452

[115] Q. Wang, P. Bu, T. P. Breckon, Unifying unsupervised domain adaptation and zero-shot visual recognition, in: 2019 International Joint Conference on Neural Networks (IJCNN), IEEE, 2019, pp. 1–8.

[116] M. Mancini, S. R. Bulo, B. Caputo, E. Ricci, Best sources forward: domain generalization through source-specific nets, in: 2018 25th IEEE international conference on image processing (ICIP), IEEE, 2018, pp. 1353–1357.

[117] J. Ryu, G. Kwon, M.-H. Yang, J. Lim, Generalized convolutional forest networks for domain generalization and visual recognition, in: International conference on learning representations, 2019, p. 0.

[118] U. Muhammad, J. Laaksonen, D. Romaissa Beddiar, M. Oussalah, Domain generalization via ensemble stacking for face presentation attack detection, International Journal of Computer Vision 132 (12) (2024) 5759–5782.

[119] J. Liang, R. He, T. Tan, A comprehensive survey on test-time adaptation under distribution shifts, International Journal of Computer Vision 133 (1) (2025) 31–64.

[120] S. Ioffe, Batch normalization: Accelerating deep network training by reducing internal covariate shift, arXiv preprint arXiv:1502.03167.

[121] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell, Tent: Fully test-time adaptation by entropy minimization, arXiv preprint arXiv:2006.10726.

[122] M. J. Mirza, J. Micorek, H. Possegger, H. Bischof, The norm must go on: Dynamic unsupervised domain adaptation by normalization, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 14765–14775.

[123] J. L. Ba, J. R. Kiros, G. E. Hinton, Layer normalization, arXiv preprint arXiv:1607.06450.

[124] Q. Wang, O. Fink, L. Van Gool, D. Dai, Continual test-time domain adaptation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 7201–7211.

[125] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, M. Tan, Efficient test-time model adaptation without forgetting, in: International conference on machine learning, PMLR, 2022, pp. 16888–16905.

[126] J. Liang, D. Hu, J. Feng, Do we really need to access the source data? source hypothesis transfer for unsupervised domain adaptation, in: International conference on machine learning, PMLR, 2020, pp. 6028–6039.

[127] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, G. E. Hinton, Adaptive mixtures of local experts, Neural computation 3 (1) (1991) 79–87.

[128] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, J. Dean, Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, arXiv preprint arXiv:1701.06538.

[129] B. Zoph, Designing effective sparse expert models, in: 2022 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), IEEE, 2022, pp. 1044–1044.

[130] B. Zoph, I. Bello, S. Kumar, N. Du, Y. Huang, J. Dean, N. Shazeer, W. Fedus, ST-MoE: Designing stable and transferable sparse expert models, arXiv preprint arXiv:2202.08906.

[131] R. Feng, N. Xi, D. Chu, R. Wang, Z. Deng, A. Wang, L. Lu, J. Wang, Y. Huang, Artemis: Autoregressive end-to-end trajectory planning with mixture of experts for autonomous driving, arXiv preprint arXiv:2504.19580.

[132] Z. Luo, W. Yang, Y. Yuan, R. Gou, X. Li, Semantic segmentation of agricultural images: A survey, Information Processing in Agriculture 11 (2) (2024) 172–186. doi:10.1016/j.inpa.2023.02.001.
URL https://www.sciencedirect.com/science/article/pii/S2214317323000112

[133] S. Tian, Y. Dong, R. Feng, D. Liang, L. Wang, Mapping mountain glaciers using an improved u-net model with cse, International Journal of Digital Earth 15 (1) (2022) 463–477.

[134] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, et al., Attention u-net: Learning where to look for the pancreas, arXiv preprint arXiv:1804.03999 1.

[135] Z. Fan, K. Liu, J. Hou, F. Yan, Q. Zang, Jaunet: A u-shape network with jump attention for semantic segmentation of road scenes, Applied Sciences 13 (3) (2023) 1493.

[136] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, K. H. Maier-Hein, nnu-net: a self-configuring method for deep learning-based biomedical image segmentation, Nature methods 18 (2) (2021) 203–211.

[137] G. Liu, L. Bai, M. Zhao, H. Zang, G. Zheng, Segmentation of wheat farmland with improved u-net on drone images, Journal of applied remote sensing 16 (3) (2022) 034511–034511.

[138] D. Karimi, S. K. Warfield, A. Gholipour, Transfer learning in medical image segmentation: New insights from analysis of the dynamics of model parameters and learned representations, Artificial intelligence in medicine 116 (2021) 102078.

[139] O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, X. Yang, P.-A. Heng, I. Cetin, K. Lekadir, O. Camara, M. A. G. Ballester, et al., Deep learning techniques for automatic mri cardiac multi-structures segmentation and diagnosis: is the problem solved?, IEEE transactions on medical imaging 37 (11) (2018) 2514–2525. doi:10.1109/TMI.2018.2837502.

[140] J. Schlemper, O. Oktay, M. Schaap, M. Heinrich, B. Kainz, B. Glocker, D. Rueckert, Attention gated networks: Learning to leverage salient regions in medical images, Medical image analysis 53 (2019) 197–207.

[141] M. Amiri, R. Brooks, H. Rivaz, Fine tuning u-net for ultrasound image segmentation: which layers?, in: MICCAI Workshop on Domain Adaptation and Representation Transfer, Springer, 2019, pp. 235–242.

[142] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2018, pp. 7132–7141.

[143] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, Q. Hu, Eca-net: Efficient channel attention for deep convolutional neural networks, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2020, pp. 11534–11542.

[144] Q. Du, H. Yang, Similarity-based unsupervised band selection for hyperspectral image analysis, IEEE geoscience and remote sensing letters 5 (4) (2008) 564–568.

[145] R. B. Lyngdoh, A. S. Sahadevan, T. Ahmad, P. S. Rathore, M. Mishra, P. K. Gupta, A. Misra, Avhyas: A free and open source qgis plugin for advanced hyperspectral image analysis, in: 2021 International Conference on Emerging Techniques in Computational Intelligence (ICETCI), IEEE, 2021, pp. 71–76.

[146] M. Younis, Investigation of Hyperspectral Data Unmixing and Its Impact on the Task of Anomaly Detection, The University of Memphis, 2023.

[147] D. Manolakis, D. Marden, G. A. Shaw, et al., Hyperspectral image processing for automatic target detection applications, Lincoln laboratory journal 14 (1) (2003) 79–116.

[148] A. M. Baldridge, S. J. Hook, C. Grove, G. Rivera, The aster spectral library version 2.0, Remote sensing of environment 113 (4) (2009) 711–715.

[149] S. K. Meerdink, S. J. Hook, D. A. Roberts, E. A. Abbott, The ecostress spectral library version 1.0, Remote Sensing of Environment 230 (2019) 111196.

[150] V. Chandola, A. Banerjee, V. Kumar, Anomaly detection: A survey, ACM computing surveys (CSUR) 41 (3) (2009) 1–58.

[151] R. C. Gonzalez, Digital image processing, Pearson education india, 2009.

[152] A. K. Jain, Fundamentals of digital image processing, Prentice-Hall, Inc., 1989.

[153] G. K. Wallace, The jpeg still picture compression standard, Communications of the ACM 34 (4) (1991) 30–44.

[154] C. Tomasi, R. Manduchi, Bilateral filtering for gray and color images, in: Sixth international conference on computer vision (IEEE Cat. No. 98CH36271), IEEE, 1998, pp. 839–846.

[155] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE transactions on image processing 13 (4) (2004) 600–612.

[156] A. A. Michelson, Studies in optics, Courier Corporation, 1995.

[157] J. R. Rice, The algorithm selection problem, in: Advances in computers, Vol. 15, Elsevier, 1976, pp. 65–118.

[158] K. A. Smith-Miles, Cross-disciplinary perspectives on meta-learning for algorithm selection, ACM Computing Surveys (CSUR) 41 (1) (2009) 1–25.

[159] A. G. Barto, Reinforcement learning: An introduction. by richard's sutton, SIAM Rev 6 (2) (2021) 423.

[160] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, Imagenet: A large-scale hierarchical image database, in: 2009 IEEE conference on computer vision and pattern recognition, IEEE, 2009, pp. 248–255.

[161] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, L. Fei-Fei, ImageNet Large Scale Visual Recognition Challenge, International Journal of Computer Vision (IJCV) 115 (3) (2015) 211–252. doi:10.1007/s11263-015-0816-y.

[162] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, arXiv preprint arXiv:1711.05101 1.

[163] D. R. Cox, The regression analysis of binary sequences, Journal of the Royal Statistical Society Series B: Statistical Methodology 20 (2) (1958) 215–232.

[164] J. R. Quinlan, C4. 5: programs for machine learning, Elsevier, 2014.

[165] L. Breiman, Random forests, Machine learning 45 (1) (2001) 5–32. doi:10.1023/A:1010933404324.

[166] T. Fushiki, Estimation of prediction error by using k-fold cross-validation, Statistics and Computing 21 (2011) 137–146.

[167] D. Anguita, L. Ghelardoni, A. Ghio, L. Oneto, S. Ridella, et al., The'k'in k-fold cross validation., in: ESANN, 2012, pp. 441–446.

[168] T.-T. Wong, P.-Y. Yeh, Reliable accuracy estimates from k-fold cross validation, IEEE Transactions on Knowledge and Data Engineering 32 (8) (2019) 1586–1594.

[169] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80–83.
URL http://www.jstor.org/stable/3001968

[170] M. Friedman, The use of ranks to avoid the assumption of normality implicit in the analysis of variance, Journal of the american statistical association 32 (200) (1937) 675–701.

[171] E. R. DeLong, D. M. DeLong, D. L. Clarke-Pearson, Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach, Biometrics (1988) 837–845.

[172] W. J. Krzanowski, D. J. Hand, ROC curves for continuous data, Chapman and Hall/CRC, 2009.

[173] S. (https://stats.stackexchange.com/users/22311/sycorax), Statistical significance (p-value) for comparing two classifiers with respect to (mean) roc auc, sensitivity and specificity, Cross Validated, , uRL:https://stats.stackexchange.com/q/358598 (version: 2018-08-01). arXiv:https://stats.stackexchange.com/q/358598.
URL https://stats.stackexchange.com/q/358598

[174] D. Chicco, G. Jurman, The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation, BMC genomics 21 (2020) 1–13.

[175] W. Kong, B. Liu, X. Bi, C. Yu, X. Li, Y. Chen, Hypersl: A spectral foundation model for hyperspectral image interpretation, IEEE Transactions on Geoscience and Remote Sensing.

[176] Y. Ganin, V. Lempitsky, Unsupervised domain adaptation by backpropagation, in: International conference on machine learning, PMLR, 2015, pp. 1180–1189.

[177] A. G. Khoee, Y. Yu, R. Feldt, Domain generalization through meta-learning: a survey, Artificial Intelligence Review 57 (10) (2024) 285.

[178] I. Loshchilov, F. Hutter, Sgdr: Stochastic gradient descent with warm restarts, arXiv preprint arXiv:1608.03983.

[179] W. Fedus, B. Zoph, N. Shazeer, Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, Journal of Machine Learning Research 23 (120) (2022) 1–39.

[180] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, et al., Mixed precision training, arXiv preprint arXiv:1710.03740.

[181] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, A. Smola, A kernel two-sample test, The journal of machine learning research 13 (1) (2012) 723–773.

[182] W. Liu, X. Wang, J. Owens, Y. Li, Energy-based out-of-distribution detection, Advances in neural information processing systems 33 (2020) 21464–21475.

[183] J. Chen, J. Mei, X. Li, Y. Lu, Q. Yu, Q. Wei, X. Luo, Y. Xie, E. Adeli, Y. Wang, et al., Transunet: Rethinking the u-net architecture design for medical image segmentation through the lens of transformers, Medical Image Analysis 97 (2024) 103280.

[184] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, Imagenet: A large-scale hierarchical image database, in: 2009 IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248–255. doi:10.1109/CVPR.2009.5206848.

[185] K. He, X. Zhang, S. Ren, J. Sun, Identity mappings in deep residual networks, in: Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part IV 14, Springer, 2016, pp. 630–645.

[186] S. Sun, J. Pang, J. Shi, S. Yi, W. Ouyang, Fishnet: A versatile backbone for image, region, and pixel level prediction, Advances in neural information processing systems 31.

[187] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, in: International Conference on Learning Representations, 2021, p. 0.
URL https://openreview.net/forum?id=YicbFdNTTy

[188] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18, Springer, 2015, pp. 234–241.

[189] A. Odena, V. Dumoulin, C. Olah, Deconvolution and checkerboard artifacts, Distill 1 (10) (2016) e3.

[190] W. Shi, J. Caballero, F. Huszár, J. Totz, A. P. Aitken, R. Bishop, D. Rueckert, Z. Wang, Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 1874–1883.

[191] G. King, L. Zeng, Logistic regression in rare events data, Political analysis 9 (2) (2001) 137–163.

[192] S. Jadon, A survey of loss functions for semantic segmentation, in: 2020 IEEE conference on computational intelligence in bioinformatics and computational biology (CIBCB), IEEE, 2020, pp. 1–7.